Number Sequences & Patterns

Year 4 · **Lesson Worksheet**

Worked Examples

A) Easy: Find the next two terms: 4, 8, 12, 16, __, __.

Answer: Add 4 each time. Next terms: 20, 24.

B) Standard: Find the missing term: 5, 10, 15, __, 25.

Answer: The rule is add 5 each time, so the missing term is 20.

C) Advanced: A pattern of counters has 3 counters in step 1, 6 in step 2, 9 in step 3. How many counters are in step 5?

Answer: The pattern goes up by 3 each step: 3, 6, 9, 12, 15. Step 5 has 15 counters.

Easy / Standard (15 questions)

1. Find the next two terms: 3, 6, 9, 12, __, __.
2. Find the next two terms: 10, 20, 30, 40, __, __.
3. Which number continues the sequence 2, 4, 6, 8, __? A) 9 B) 10 C) 11 D) 12
4. Find the missing term: 5, __, 15, 20, 25.
5. Find the next two terms: 50, 45, 40, __, __.
6. Find the next two terms: 1, 3, 5, 7, __, __.
7. Select ALL the sequences that count up in steps of 5: A) 5,10,15,20 B) 2,4,6,8 C) 15,20,25,30 D) 3,6,9,12 E) 30,35,40,45
8. Find the missing term: 4, 8, __, 16, 20.
9. Find the next two terms: 100, 90, 80, __, __.
10. Which of these is the rule for the sequence 6, 12, 18, 24...? A) add 4 B) add 6 C) add 8 D) multiply by 2
11. Find the next two terms: 2, 5, 8, 11, __, __.
12. Find the missing term: 30, 25, __, 15, 10.
13. Select ALL the numbers that could continue the sequence 9, 18, 27, 36...: A) 45 B) 40 C) 54 D) 50 E) 63
14. Find the next two terms: 7, 14, 21, __, __.
15. Which number is missing? 12, 16, 20, __, 28. A) 22 B) 23 C) 24 D) 25

Standard (15 questions)

1. Find the next two terms: 2, 4, 8, 16, __, __.
2. Find the next two terms: 1, 2, 4, 8, __, __.
3. What is the rule for the sequence 3, 6, 12, 24...? A) add 3 B) add 6 C) double each time D) add 12

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4. Find the missing term: 2, 6, __, 14, 18.

5. Find the next two terms: 80, 40, 20, __, __.

6. A pattern of squares has 1 square in row 1, 3 squares in row 2, 5 squares in row 3. How many squares are in row 4?

7. Select ALL the sequences below that step up by 4 each time: A) 4,8,12,16 B) 3,7,11,15 C) 5,10,15,20 D) 6,10,14,18 E) 2,5,8,11

8. Find the missing term: 9, 12, __, 18, 21.

9. Find the next two terms: 95, 85, 75, __, __.

10. Which of these is the missing term? 40, 34, 28, __, 16. A) 20 B) 22 C) 24 D) 26

11. Find the next two terms: 11, 22, 33, 44, __, __.

12. A pattern of dots doubles each time: 3, 6, 12, 24, __. What is the next term?

13. Find the missing term: 100, 85, __, 55, 40.

14. Which rule describes the sequence 1, 4, 9, 16...? A) add consecutive odd numbers B) add 3 each time C) multiply by 4 D) add 4 each time

15. Select ALL the numbers below that are square numbers: A) 9 B) 12 C) 16 D) 20 E) 25

Standard / Hard (15 questions)

1. Find the next two terms: 4, 9, 14, 19, __, __.

2. Which number continues the sequence 5, 10, 15, 20, __? A) 22 B) 24 C) 25 D) 30

3. A row of tables seats 4 people. Two tables pushed together seat 6 people (each extra table adds 2 seats). How many people can 5 tables seat?

4. Find the missing term: 3, __, 11, 15, 19.

5. Select ALL the sequences below where the difference between terms is always the same: A) 4,8,12,16 B) 1,2,4,8 C) 10,20,30,40 D) 3,5,8,12 E) 90,80,70,60

6. A pattern of triangles is made from matchsticks: 1 triangle uses 3 sticks, and each extra triangle added in a row needs 2 more sticks. How many sticks for 4 triangles?

7. Find the next two terms: 1, 4, 9, 16, __, __.

8. Which number is the odd one out — it is not a multiple of 6? A) 12 B) 18 C) 25 D) 30

9. A sequence starts at 40 and decreases by 5 each time. What is the 4th term?

10. Find the missing term: 4, 10, __, 22, 28.

11. A garden path is made from square slabs: 1 slab needs 4 edges, and each extra slab added in a row needs 3 more edges. How many edges for 5 slabs?

12. Select the TWO sequences below that double each time: A) 2,4,8,16 B) 1,3,5,7 C) 3,6,12,24 D) 4,8,12,16 E) 5,10,15,20

13. Find the next term: 60, 54, 48, 42, __.

14. Which term comes next? 3, 6, 12, 24, __. A) 30 B) 36 C) 48 D) 27

15. A function machine adds 5, then subtracts 2. If the input is 4, what is the output?

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